



Molecular dynamics simulation of the structure and properties for the CaO–SiO₂ and CaO–Al₂O₃ systems



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ABSTRACT

The microstructure and transport properties of CaO–SiO₂ and CaO–Al₂O₃ systems were studied by molecular dynamics simulations and the relations between the structure and transport properties were also discussed. The results show that the simulation results of both structure and transport properties of CaO–SiO₂ and CaO–Al₂O₃ systems are well consistent with the experimental values. With the addition of CaO content, the melt structure would be depolymerized resulting in decrease of viscosity and enhancement of electrical conductivity in both systems. The relations between the structure and transport properties were established by fitting the calculated results and those results show that the decrease of melt polymerization degree leads to decrease of viscosity and increase of electrical conductivity in both systems. Relation between the logarithm of viscosity and the logarithm of conductivity of CaO–SiO₂ system shows good agreement with the published literature. However, it needs further investigation for CaO–Al₂O₃ system.

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1. Introduction

Since the aluminosilicate melts are very important systems in the field of metallurgy, ceramics, glass, and geology, their structure and properties are of great interest for understanding their physical and transport behaviors. Although there are many experiment results about properties of aluminosilicate melts, it is still hard to give an explanation about the behavior of melts in essence. In addition, with increasing requirements that most experiments must be conducted at extreme thermodynamic conditions, it presents several problems to use conventional equipment which would be very difficult to attain and work at very high temperatures, and when this is possible there is always the risk that the sample is polluted by the material of the container. In recent years, due to the advantage of the numerical simulation allows one to avoid experimental difficulties, the molecular dynamics (MD) simulation has been widely employed as a very useful method to calculate thermodynamic, structural and transport properties for high melting temperature and high vapor pressure materials.

Up to now, the structure properties of aluminosilicate melts have already been studied systemically by various experimental and theoretical methods [1–11]. Moreover, there are also some researches [12–17] about the properties of aluminosilicate melts by MD simulation that have been reported. Belonoshko and Dubrovinsky [12,13] have successfully assessed the thermodynamic and structural properties for the silica and magnesium silicates at various temperatures and pressures by

using the transferable pairwise interatomic potential model. Takei et al. [14] have calculated the metastable immiscibility region in the Al₂O₃–SiO₂ system from the thermodynamic and structural data obtained by MD simulation. The structural and thermodynamic properties for the CaO–SiO₂ melts were calculated by Seo and Tsukihashi [15] using MD simulation, and then based on the results obtained from MD calculation, the phase diagram for the CaO–SiO₂ system, especially the liquid–liquid immiscibility region in the SiO₂-rich region was estimated. Guillot and Sator [16,17] have investigated the thermodynamic, structural and transport properties of natural silicate melts under different pressures by MD simulation, whereas there were large error bars associated with the computed values of electrical conductivities. In view of the previous papers, transport properties of aluminosilicate melts have not been well studied yet.

In this paper, the structure information, viscosity and electric conductivity of CaO–SiO₂ and CaO–Al₂O₃ systems were calculated based on MD simulation and compared with the experimental values. Moreover, since the viscosity and electrical conductivity are very sensitive to the structure change of melts, the essential relations between them were discussed. The results would have great significance in providing guidance for the research and development of aluminosilicate materials.

2. Computational methodology

2.1. Molecular dynamic simulation method

The primary job of molecular dynamics simulation is to choose an appropriate potential function and its corresponding parameters which are suitable for the calculation of particle interactions. Since the

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two-body potential function Born–Mayer–Huggins (BMH) [18], which is composed of the long-range coulomb interaction, short-range repulsion interaction and van der Waals force, has been generally and successfully used in MD simulation, it was applied here:

$$U_{ij}(r) = \frac{q_i q_j}{r_{ij}} + A_{ij} \exp(-B_{ij}r) - \frac{C_{ij}}{r_{ij}^6} \quad (1)$$

where $U_{ij}(r)$ is the interatomic pair potential, q_i and q_j are the selected charges, and in order to ensure the transferability of the interaction potential with the melt composition, the valence assigned to the atoms is usually kept fixed for all compositions, r_{ij} represents the distance between atoms i and j , A_{ij} and C_{ij} are energy parameters for the pair ij describing repulsive and van der Waals attractive forces, respectively and B_{ij} is a e-folding length characterizing the radically symmetric decay of electron repulsion energy between atom pair ij . As for the potential parameters, there are three categories including force field parameters, empirical parameters and non-empirical parameters. With the comprehensive consideration of the accuracy and computational cost, empirical values [19] which have been verified for MD simulation of aluminosilicate slags are applied in this paper and listed in Table 1.

Since the BMH potential function regards the melts as a completely ionic solution during the simulation process, the simulated system should be chosen in the completely molten composition range at the corresponding temperature. According to the phase diagrams of CaO–SiO₂ and CaO–Al₂O₃ systems [20], the molten melt compositions are determined and the total atom number of each sample is set as about 2000. The density of each sample is obtained from the experimental formula in the paper [21]. The compositions, atom numbers and density of each sample are presented in Table 2.

The MD simulations were performed with the Materials Explorer program in NVT ensemble, which means the simulations are run keeping the number of particles (N), sample volume (V) and the temperature (T) of the system constant. The equations of motions for atoms were solved with a time step of 1 fs (10^{-15} s) by the leap-frog algorithm and the data would be saved by each of the 10 steps. As the atoms are bearing charges, the long-range Coulomb forces which presented in the first term of Eq. (1) are evaluated using the Ewald sum method with a precision of 10^5 . When calculating the repulsive forces, the cutoff of potential was set to 10 Å, a good trade-off between accuracy and computational cost. Meanwhile, the pressure of system is controlled by the Parrinello and Rahman method. In the simulation process, an appropriate number of atoms of each type are placed in the primary MD cell with a random initial state and then the volume of the cell is decided by the atom numbers and the density. Since the number of atoms in the real world is always far more than the quantities that the computer can support, the periodic boundary conditions are applied on all sides of the model box to create an infinite system with no boundaries so that the calculation results would be more convincing.

Through a lot of preliminary explorations, a stable simulation process was determined as follows for each sample. At the beginning of the simulation, the initial temperature was fixed at 4000 K for 24,000 steps to mix the system completely and eliminate the effect of

Table 2

The compositions, atomic numbers and density of each sample at 1873 K.

Groups	Mole fraction, %			Atomic number					Density, g/cm ³ 1873 K
	CaO	SiO ₂	Al ₂ O ₃	Ca	Si	Al	O	Total	
CS1	40	60	/	308	462	/	1232	2002	2.4705
CS2	45	55	/	353	431	/	1215	1999	2.5261
CS3	50	50	/	400	400	/	1200	2000	2.5846
CS4	55	45	/	449	367	/	1183	1999	2.6462
CA3	50	/	50	286	/	571	1143	2000	2.5593
CA4	55	/	45	328	/	537	1134	1999	2.6063
CA5	60	/	40	375	/	500	1125	2000	2.6581
CA6	65	/	35	426	/	459	1115	2000	2.7154
CA7	70	/	30	483	/	414	1103	2000	2.7793

the initial distribution. Then, the temperature was cooled down to 1873 K through 96,000 steps and the temperature reduction is accomplished by direct velocity rescaling of each molecule. After equilibrium calculation, the system is relaxed for another 60,000 steps. Finally, structure information and transport properties of melts could be calculated and analyzed.

2.2. Statistics of structure information

The nearest neighbor distributions, which are determined by statistical analysis of atom locations using radial distribution functions (RDFs), could encapsulate short-range order [22] in melts and enable the connection between atomic structure and macro-properties to be understood. The equation of RDF is expressed as below.

$$g_{ij}(r) = \frac{V}{N_i N_j} \sum_j \frac{\langle n_{ij}(r - \Delta r/2, r + \Delta r/2) \rangle}{4\pi r^2 \Delta r} \quad (2)$$

In addition, the coordination numbers (CNs) for atom i around atom j are determined by numerical integration of the RDFs within the cut-off radius, which corresponds to the first minima of the corresponding partial $g_{ij}(r)$.

$$N_{ij}(r) = 4\pi \frac{N_j}{V} \int_0^r g_{ij}(r) r^2 dr \quad (3)$$

where, V is the volume of the MD primary cell and N is the number of particles. $n_{ij}(r - \Delta r/2, r + \Delta r/2)$ is the average number of atom j surrounding a central i atom within a defined cut-off distance $r \pm \Delta r/2$. The RDFs give the probability of finding an ion within a distance Δr from a specified particle at the location of r and the CNs show the average coordination number for atom i around atom j .

As for the concentration of oxygen types and the distribution of structure units Q^n , these structure information would be counted by Matlab programming based on the atomic coordinates in space which derived from MD simulation.

2.3. Calculation of transport properties

Through the statistical analysis of the particle trajectory by means of MD simulation, the function of the mean square displacement (MSD) about time would be generated.

$$MSD = \langle \Delta \bar{r}(t)^2 \rangle = \frac{1}{N} \left\langle \sum_i |r_{i(t)} - r_{i(0)}|^2 \right\rangle \quad (4)$$

where, $r_{i(t)}$ presents the location of atom i at the time of t and the bracket expresses an ensemble average taken over many origin times. Combined with the statistical mechanics and statistical thermodynamics, some transport properties of melts such as self-diffusion coefficient,

Table 1

The potential parameters of CaO–SiO₂ and CaO–Al₂O₃ systems used in this paper.

Atom1	Atom2	A_{ij} (eV)	B_{ij} (1/Å)	C_{ij} (eV·Å ⁶)
O	O	1,497,049	5.882353	17.34
O	Al	86,057.58	6.060606	0
O	Si	62,794.37487	6.060606	0
O	Ca	717,827	6.060606	8.67
Ca	Ca	329,051.6	6.25	4.335
Ca	Si	26,674.6764	6.25	0
Ca	Al	36,918.57	6.25	0
Si	Si	2,162.3914	6.25	0
Al	Al	4,142.149	6.25	0

viscosity and electric conductivity would be calculated on account of the MSD curves of particles. The relation between the self-diffusion coefficient D and the MSD curve is as shown below [23].

$$D = \lim_{t \rightarrow \infty} \frac{1}{6} \frac{d[\Delta \bar{r}(t)^2]}{dt} \quad (5)$$

Then the viscosity η and electric conductivity σ can be obtained by combining the self-diffusion coefficient D with the Einstein–Stokes formula and Nernst–Einstein equation respectively.

$$\eta = \frac{K_B T}{D \lambda} \quad (6)$$

$$\sigma = D \frac{n q^2}{K_B T} \quad (7)$$

where, K_B is the Boltzmann constant which equals to 1.38×10^{-23} J/K, T is the temperature of the system, λ is the step of ion diffusion and it is generally considered equaling to the diameter of ion ($\lambda = 2R$), η presents the unit volume concentration of the carrier atom which equals to the number of the carrier atom in a unit volume, and q is the charge of ion.

Based on the calculation results, the connections between structure information and transport properties were established and discussed, and these could help us understand the action mechanism of molten melts, so as to give guidance for the research and development of aluminosilicate materials.

3. Results

3.1. RDFs and CNs

The abscissa value corresponding to the first peak of RDF is the average bond length of first neighbors. With the integration of RDF, the CN would be obtained and the ordinate value corresponding to the platform of CN is considered to be the average coordination number. The results could be compared with experimental data to verify the accuracy of MD simulation.

In Fig. 1 is presented the RDFs and CNs of Si–O and Al–O in CaO–SiO₂ and CaO–Al₂O₃ systems at 1873 K while the content of CaO equals to 50 mol% as an example. Since the abscissa values corresponding to the first minimums of $g_{Si-O}(r)$ and $g_{Al-O}(r)$ are around 2.0 Å and 2.5 Å respectively, the cutoff radiuses (2.0 Å for Si and 2.5 Å for Al) were therefore decided.

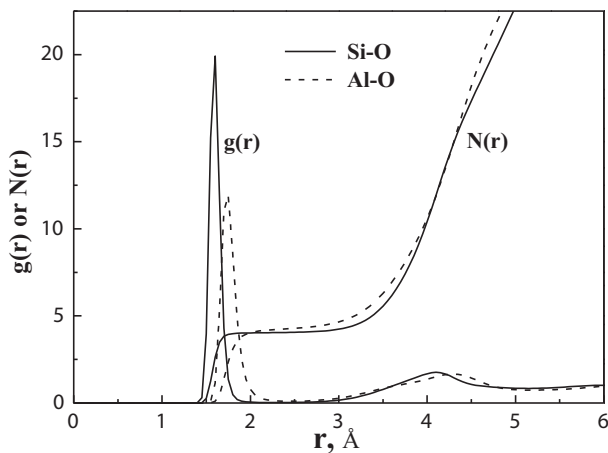


Fig. 1. The RDFs and CNs of Si–O and Al–O in CaO–SiO₂ and CaO–Al₂O₃ systems while CaO = 50 mol% at 1873 K.

3.2. Distribution of oxygen types and structure units Q^n

During the simulation process, accurate coordinates in space of every atom in the cubic model box were saved to analyze the distances between atoms so as to get the distribution of oxygen types and structural units Q^n by Matlab programming. It is commonly considered that the oxygen in traditional silicate network structure can be divided into bridge oxygen (O_b), non-bridge oxygen (O_{nb}) and free oxygen (O_f), while oxygen triclusters [24–28] occurred to balance the charge in alumina containing systems. Two atoms (e.g Si–O) are connected if the distance between the two atoms is less than the aforementioned cutoff radius (2.0 Å for Si and 2.5 Å for Al). According to the numbers of Si or Al that each particular oxygen atom coordinated with, all kinds of oxygen types have been distinguished and counted in CaO–SiO₂ and CaO–Al₂O₃ systems, respectively.

A detailed proportion analysis of oxygen types is given in Fig. 2 as function of composition in CaO–SiO₂ and CaO–Al₂O₃ systems at 1873 K. For CaO–SiO₂ system, Fig. 2(a) exhibits the simulation results of the relative amount of three different oxygen anions and the Raman experimental results of Tsunawaki [29] at 1873 K. As for distribution of the structure units Q^n , according to the numbers of O_b and oxygen tricluster in one tetrahedron, the proportion of Q^n have been analyzed and presented in Fig. 3.

3.3. Viscosity and electrical conductivity

As mentioned before, by combing the MSD curve which was produced by MD simulation and Eq. (5), the self-diffusion coefficient of O

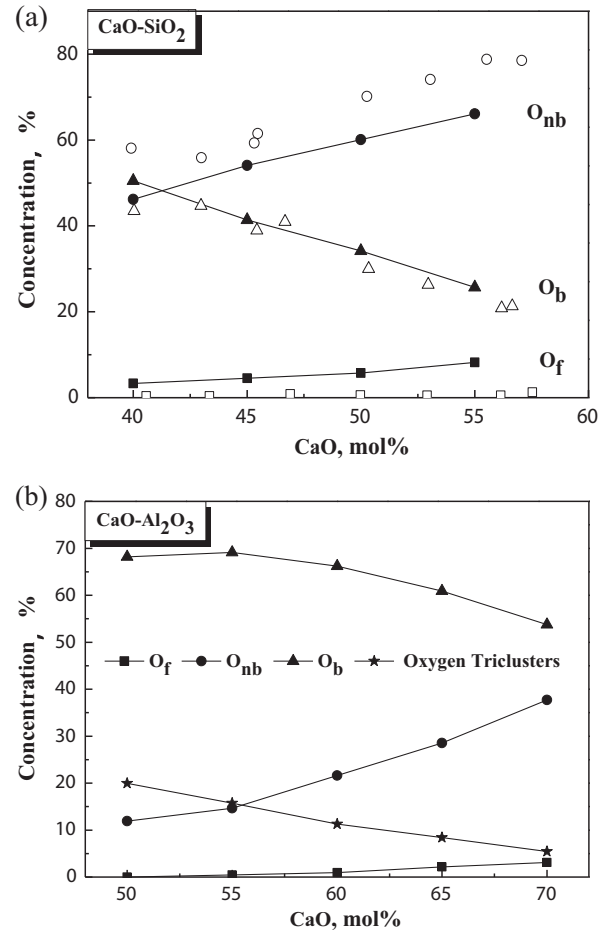


Fig. 2. The distribution of oxygen types in (a) CaO–SiO₂ and (b) CaO–Al₂O₃ systems at 1873 K.

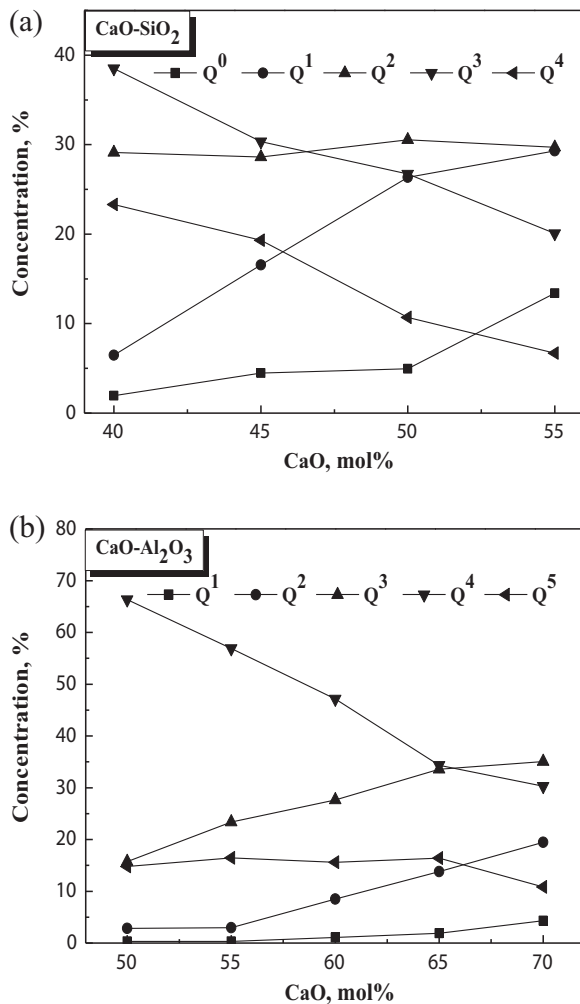


Fig. 3. The distribution of structural units Q^n in (a) CaO-SiO₂ and (b) CaO-Al₂O₃ systems at 1873 K.

ions could be gained. Then the viscosity of each sample was calculated according to the Einstein-Stokes formula, thereinto the ion diffusion length λ equals to the diameter of O ion 2.8 Å. The viscosities changed with CaO content and compared with the measured values in CaO-SiO₂ [30–34] and CaO-Al₂O₃ [35–37] systems at various temperatures are presented in Fig. 4.

As for the electric conductivity, it can be calculated by combing the diffusion coefficient of Ca ion and the Nernst-Einstein equation. Fig. 5 demonstrates that the electric conductivities changed with CaO content and compared with experiment values measured by many scholars in CaO-SiO₂ [38–46] and CaO-Al₂O₃ [47–54] systems at 1873 K.

4. Discussion

4.1. Structure properties

From Fig. 1, it can be seen that, the abscissa values of first peaks of $g_{Si-O}(r)$ and $g_{Al-O}(r)$, which represent the average bond length of the corresponding two atoms, are equal to 1.612 Å and 1.75 Å, respectively. These values have an exceptionally high level of agreement with the nearest neighbor distances of Si-O and Al-O bonds presented in other literatures [55–65]. Since the average distance of Al-O is greater than that of Si-O, and the first peak value of $g_{Al-O}(r)$ is much smaller than that of $g_{Si-O}(r)$, these suggest that the bonding ability of Al-O is weaker than Si-O, meaning that the Al-O network is not as stable as the Si-O network. In addition to the first peaks, there are gentler second

peaks on Si-O and Al-O RDFs, indicating that long-range ordered structure appeared with increasing distance between two atoms. The running CNs also help to promote an understanding of the balance of attraction and repulsion between the particles. The ordinate values corresponding to smooth platforms on each $N_{Si-O}(r)$ and $N_{Al-O}(r)$ equal to around 4, meaning that the overwhelming majority of Si and Al atoms are 4-coordinated with oxygen. However, through the comparison, it can be found that the platform of the Al-O CN curve is a bit wider and gentler than that of Si-O CN curve, demonstrating that the stability of SiO₄ tetrahedron structure is better than that of AlO₄ tetrahedron structure.

In Fig. 2(a), it can be seen that, in CaO-SiO₂ system, with the increase of CaO, the proportion of O_{nb} increases dramatically and the ratio of O_b reduces greatly, while there are a relatively small number of O_f presenting a small amplitude growth trend. This is due to the fact that CaO acts as a network modifier and the O^{2-} dissociated from CaO would bring about depolymerization of the melt structure. From comparison, the difference of simulation results and experiment values is limited to 12%. To be specific, the proportion of O_{nb} obtained from simulation is less than the experiment results while the content of O_b is very consistent with the experiment values. Moreover, up to 8% free oxygen O_f exists in the simulation results, but this concentration is hard to detect in the experiment. So it is very likely that the O_{nb} concentration obtained from experiment may include the content of O_f and thus it seems to be

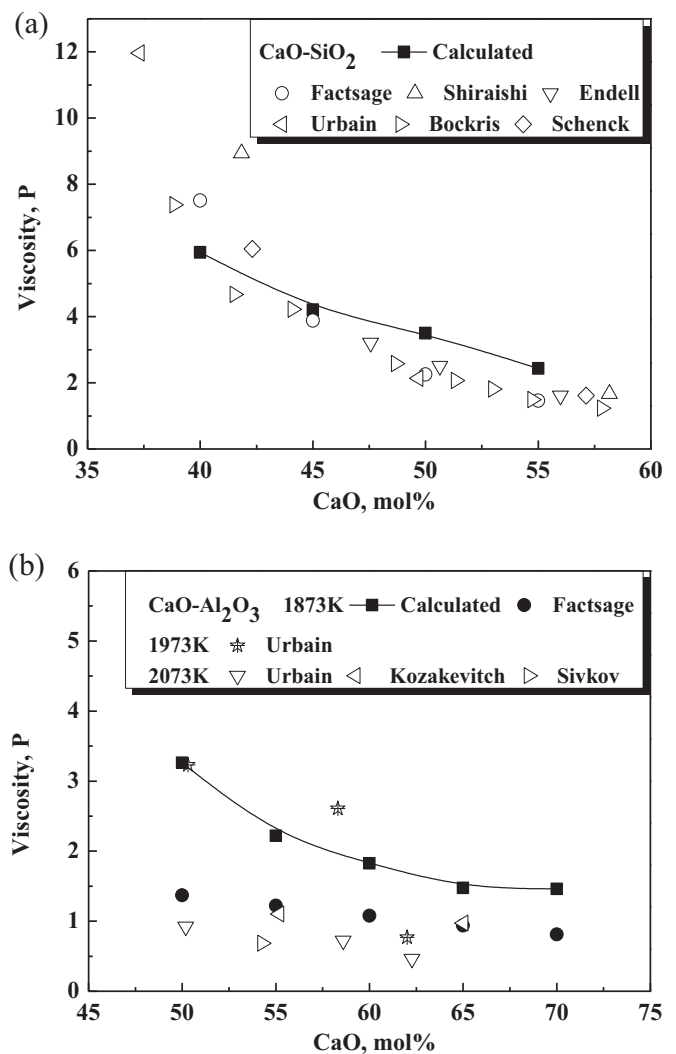


Fig. 4. The viscosities changed with CaO content and compared with the measured values in (a) CaO-SiO₂ and (b) CaO-Al₂O₃ systems at 1873 K.

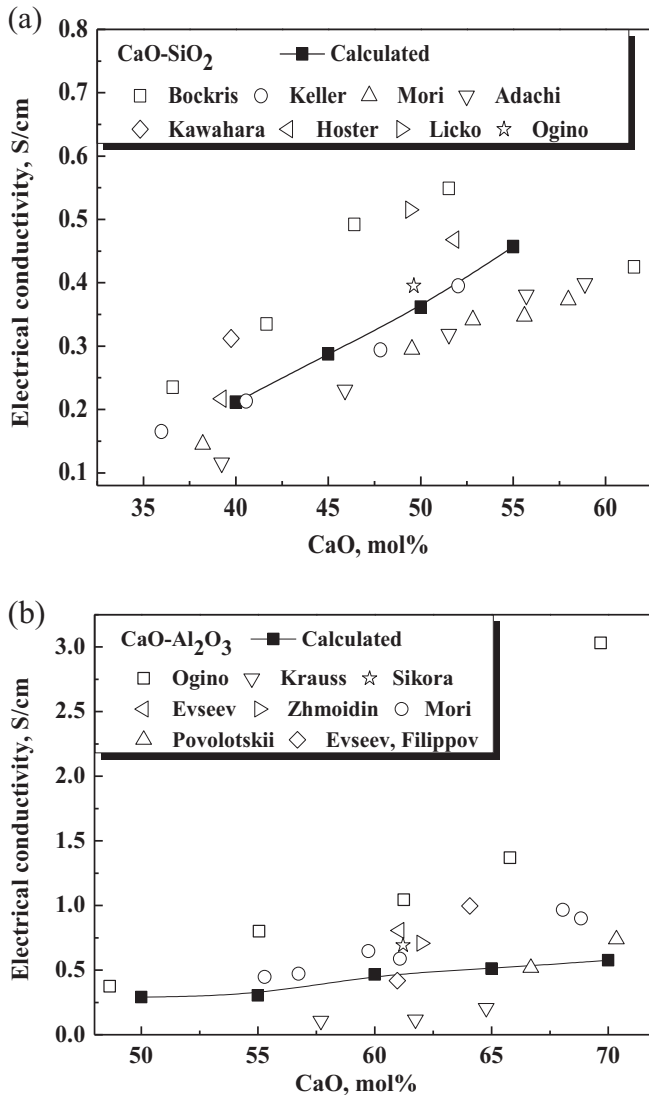


Fig. 5. The electric conductivities changed with CaO content and compared with the measured values in (a) CaO–SiO₂ and (b) CaO–Al₂O₃ systems at 1873 K.

reasonable that there are inconsistencies between simulation and experiment results. In addition, the phenomenon in CaO–Al₂O₃ system shown in Fig. 2(b) is similar to that in CaO–SiO₂ system that increasing the CaO content, the melt structure becomes simple with O_b and oxygen tricluster breaking down into O_{nb} and O_f. According to MD simulation results of calcium aluminate glasses at 300 K by Kang [65], when the content of CaO is 50 mol%, the O_{nb} ratio is 11.6% while the O_b proportion is 88.8%, it is very similar to the results in our calculation which show that the O_{nb} ratio is 11.9% while the proportion of the sum of O_b and oxygen tricluster is 88.1%. However, the simulation temperatures are different between in this paper and in Kang's, and in Kang's simulation results, there are no oxygen triclusters analyzed, so it is just not sufficient to verify the simulation results of CaO–Al₂O₃ system by comparing with Kang's results, thus further studies are needed.

As can be seen from Fig. 3, in CaO–SiO₂ system, the concentration of structure units Q⁰, Q¹ increase and Q³, Q⁴ decrease while Q² changes small with the increase of CaO content, this is because addition of the network modifier content leads to depolymerization reaction that complex structural units break down into simple units in melts. As for CaO–Al₂O₃ system, though the melt structure is much more complex than that of CaO–SiO₂ system, the increase of CaO content also simplifies the melt structure that the structural units Q⁵ and Q⁴ depolymerize to Q³, Q² and Q¹. The simulation results trends in both systems are consist

with the results in literatures [58,62,65] while there are some numerical differences. It would be mainly due to the fact that the disorder of glasses measured by experiment and that of melts calculated by MD simulation are not the same.

4.2. Transport properties

The viscosity of melts is very structure dependent that change sharply as the melt structure changes. As illustrated in Fig. 4, with the increase of CaO content, the viscosities of samples in both CaO–SiO₂ and CaO–Al₂O₃ systems decrease which leads to an improvement of melt liquidity. The functional mechanism of CaO on melt viscosity would be explained below, taking CaO–SiO₂ system for an example. The acidic oxide SiO₂ forms Si–O tetrahedron in melts and by sharing the oxygen ions, these tetrahedrons are joined together in chains or rings to form a network structure. When the basic oxide CaO is introduced, it will release oxygen ions to combine with the Si cation in the Si–O tetrahedron so as to break the network structure. Consequently, the viscosity, which reflects the viscous resistance of the melt in the flow process and prominently relies on the big complex anions, would decrease in melts with simple structure units. Likewise, the increase of CaO content in CaO–Al₂O₃ system would destroy the Al–O network which reduces the viscosity as a consequence. Meanwhile, the MD simulation results are consistent with experimental results, especially in CaO–SiO₂ system. Since there are not enough experimental values of viscosity in CaO–Al₂O₃ system at 1873 K, the experiment results of viscosity at 1973 K and 2073 K which are smaller than simulation results at 1873 K would partly and indirectly prove the simulation accuracy. In sum, owing to the fact that the calculated results are in good agreement with measured values, the conclusion can be drawn that the MD simulation results are accurate and reliable in this paper.

Concerning the electrical conductivity, it is mainly influenced by two factors that the transport resistance and the concentration of metal cation. The transport resistance is mainly determined by the polymerization degree of melt structure and the size of transporter. Generally, the higher the polymerization degree, the greater the resistance, thereby the electrical conductivity becomes lower. As for the cation concentration, as a general rule, electrical conductivity increases with the enhancement of the concentration. As it can be seen in Fig. 5, with the increase of CaO content, the electrical conductivities in both CaO–SiO₂ and CaO–Al₂O₃ systems increase, this is because the addition of CaO will on the one hand bring about a decrease of transfer resistance by releasing oxygen ions to depolymerize the network structure, and on the other hand, result in an increase of the metal cation concentration, and both of them benefit the enhancement of conductivity. Since the calculated results are in accord with the measured values in both systems, it can be considered that the MD simulation is comparatively successful in this paper.

4.3. Relation between structure and transport properties

The degree of de-polymerization of melt is frequently represented by the parameter NBO/T. NBO is the number of non-bridging oxygen which is characterized as O_{nb} in this paper, and T is the number of network former (number of Si atoms in CaO–SiO₂ system and number of Al atoms in CaO–Al₂O₃ system). Based on the linear fitting method, the functions of viscosities and electric conductivities of CaO–SiO₂ and CaO–Al₂O₃ systems on the de-polymerization degree parameter NBO/T are obtained and indicated in Fig. 6.

The de-polymerization degree represents the complexity of the melt system that the greater the NBO/T, the simpler the melt structure, thus the viscosity and electric conductivity of the corresponding melt would be described. From Fig. 6, it can be seen that, with the increase of NBO/T which means that the polymerization degree of melts decreases, the viscosities decrease and electrical conductivities increase because of the diffusivities of both structure units and Ca cations have been

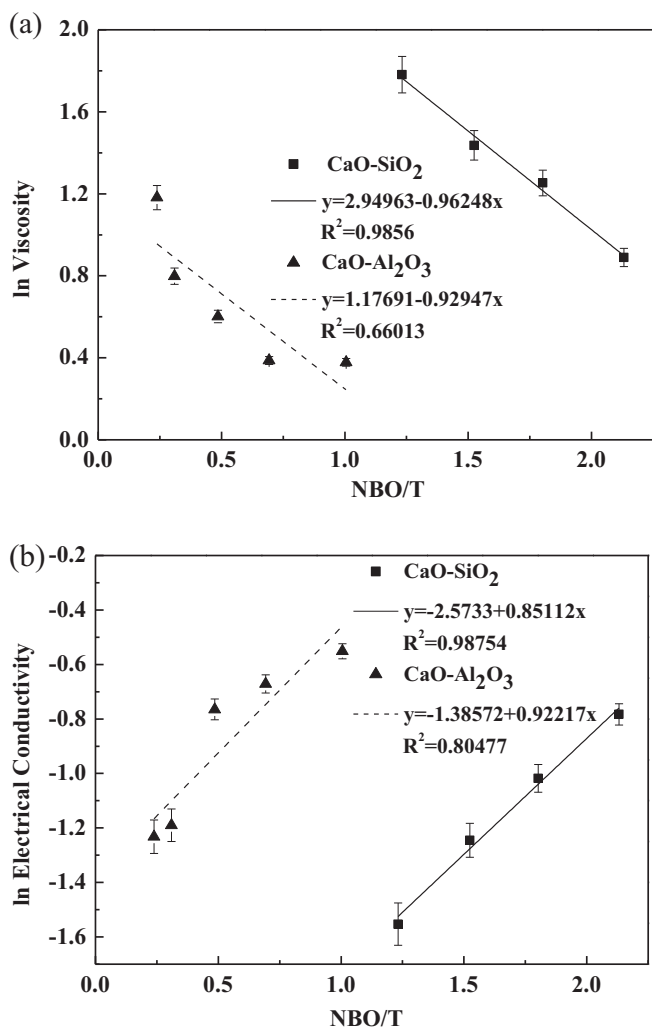


Fig. 6. The relation between (a) viscosities and (b) electrical conductivities with NBO/T in CaO–SiO₂ and CaO–Al₂O₃ systems at 1873 K.

strengthened. However, since the structure of CaO–Al₂O₃ melt system is more complicated due to the amphoteric property of Al₂O₃ and the charge compensation in Al–O tetrahedron, the value of coefficient of determination R^2 for CaO–Al₂O₃ system is relatively low, and it is not enough to discuss the relation between structure and properties only based on the parameter NBO/T, so it needs further investigation for CaO–Al₂O₃ system.

Fig. 7 shows the relation between viscosity and electrical conductivity of binary silicate melts (MO–SiO₂ systems) studied by Zhang [66] that the logarithm of viscosity varied linearly with the logarithm of conductivity ($\ln \eta = 0.15 - 1.10 \ln \sigma$). As it can be seen that the solid squares which represent the MD simulation results of CaO–SiO₂ system are well close to the straight line, it can be concluded that the MD simulation results of viscosity and electrical conductivity for CaO–SiO₂ systems are quite convincing.

5. Conclusions

Molecular dynamic simulation was carried out on CaO–SiO₂ and CaO–Al₂O₃ systems to study the structure and transport properties of these systems, and the results are as follows.

- (1) The MD simulation results of structures of CaO–SiO₂ and CaO–Al₂O₃ systems are well consistent with experimental values and the addition of CaO content would depolymerize the melt

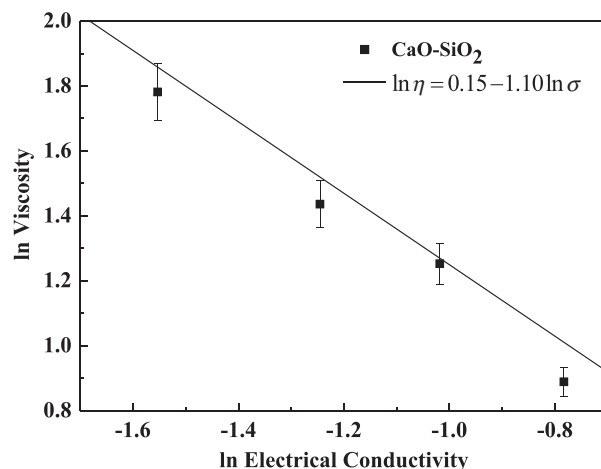


Fig. 7. The relation between the logarithm of viscosity and the logarithm of conductivity of CaO–SiO₂ system.

structure of both systems. However, it needs further investigation in CaO–Al₂O₃ system.

- (2) The calculated transport properties of CaO–SiO₂ and CaO–Al₂O₃ systems at 1873 K are in good agreement with the observed results, and with the increase of CaO content, the viscosity of melts decreased while the electrical conductivity was enhanced in both systems.
- (3) The relation between the structures and transport properties is established by fitting the calculated results. It shows that the increase of NBO/T leads to decrease of viscosities and increase of electrical conductivity in both CaO–SiO₂ and CaO–Al₂O₃ systems. Relation between the logarithm of viscosity and the logarithm of conductivity of CaO–SiO₂ system shows good agreement with the published literature.

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